Nahian Tahmin, Ph.D.

Machine Learning Engineer | AI Engineer | Applied Scientist

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PROFESSIONAL SUMMARY

Machine learning engineer with a Ph.D. in Data Science and Engineering and a B.Eng. in Software Engineering. Designs, trains, and evaluates deep learning models end to end in Python and PyTorch: multimodal Transformer architectures, large language model (LLM) fine-tuning, and explainable AI tooling on high-dimensional data in the cloud. Builds reproducible ML pipelines with strong baselines, ablations, and validation on independent data. Grounded in object-oriented programming, data structures and algorithms, Agile, and DevOps.

EDUCATION

University of Tennessee, Knoxville Aug 2022 - Aug 2026

Ph.D., Data Science and Engineering. GPA 3.95/4.00.

Monash University Malaysia Oct 2017 - Oct 2021

B.Eng., Software Engineering.

PUBLICATIONS

Tahmin N, Chinthala L, Marsico F, Buonaiuto S, Mohammed A, Carlisle A, Gautam Y, Colonna V, Mersha T, Davis RL, Khojandi A. Machine learning models incorporating genotype and ancestry improve severe asthma risk prediction. *Scientific Reports* 15, 40243 (2025). doi.org/10.1038/s41598-025-24080-x

Tahmin N, Chinthala LK, Mersha TB, Davis RL, Khojandi A. X-Admix: An Interpretable Multimodal Cross-Attention Framework for Integrating Genotype, Local Ancestry, and Social Drivers of Health in Admixed African American Populations. *medRxiv* preprint (2026); under review at PLOS Computational Biology. doi.org/10.64898/2026.09.03.26362179

Pokhrel D, **Tahmin N**, Carlisle A, Mersha T, Davis RL, Khojandi A. Assessing the Role of Social Determinants in Asthma Across Racial Groups: A Comparative Model Analysis. *IEEE SIEDS* (2025). doi.org/10.1109/SIEDS65500.2025.11021139

Predicting Precision Inhaled Corticosteroids Response in Asthma Patients. Presented at the ASHG Annual Meeting 2023, Abstract 2023-A-1763-ASHG.

Integrating genomics, local ancestry, and social drivers of health (SDOH) on asthma medication response prediction. Presented at ASHG Annual Meeting 2023 and INFORMS Annual Meeting 2024 and 2025.

PROJECTS

Multimodal Deep Learning and LLM Pipelines for High-Dimensional Prediction Aug 2022 - Jul 2026

University of Tennessee, Knoxville

- **Model architecture and training.** Designed and implemented X-Admix in PyTorch: a three-stream deep learning model with pairwise cross-attention and a learned softmax-gated fusion layer. Reached AUC 0.771 on a low-signal, high-dimensional prediction task where tuned classical machine learning baselines stayed near chance (AUC 0.566 or lower), and the trend held on a second, independent dataset roughly 3x larger.
- **LLM fine-tuning and input design (ongoing).** Benchmarked four serialization schemes for feeding structured tabular data to pretrained language models under a fixed token budget. Compressed encodings beat verbose prose (AUC 0.71 vs. 0.63) by fitting far more records into the context window. Designed an embedding-injection scheme that decouples information per record from token cost and extends to unseen inputs without adding parameters.
- **Model evaluation and explainability.** Built an attribution system combining feature ablation, gradient-based attribution (Captum), and SHAP, with a leave-one-component-out ablation protocol and a consensus ranking based on cross-fold reproducibility.
- **ML pipelines for high-dimensional data.** Built feature engineering, feature selection, and ensemble pipelines (Lasso, RFE, Elastic Net, Random Forest, stacking) for data with about 5 million features and 248 samples; the stacked model beat single-source baselines by 5 AUC points. Published in Scientific Reports (2025).
- **Cloud and GPU training.** Ran data engineering and GPU model training on Google Cloud and the All of Us Researcher Workbench with PySpark, experiment tracking in Weights and Biases, and version control in Git.

Advisor, Vertically Integrated Program (VIP) Team, Criminal Sentences Project, *Human-Augmented Analytics Group*

- Advised a cross-functional student team building Named Entity Recognition (NER) natural language processing models on federal court records. Supervised evaluation criteria and error analysis, and mentored the team on data preparation, NER training and validation, and LLM-based labeling automation.

Social Determinants in Asthma Across Racial Groups: A Comparative Model Analysis

- Mentored a junior Ph.D. student to build interpretable machine learning models with targeted ablations; oversaw end-to-end study design. Published at IEEE SIEDS 2025.

SKILLS

Programming Languages: Python, SQL, Java, C, C++, R, MATLAB
Machine Learning and Deep Learning: PyTorch, TensorFlow, Transformers, Hugging Face, Large Language Models (LLM), fine-tuning, PEFT/LoRA, scikit-learn, Pandas, NumPy
Evaluation and Explainability: Captum, SHAP, ablation studies, cross-validation, hyperparameter tuning, Weights and Biases
Data and Infrastructure: PySpark, Google Cloud, Google Cloud Storage, GPU training, Oracle, Git/GitHub, Power BI
Software Engineering: Object-oriented programming (OOP), data structures and algorithms, Agile methodology, software quality assurance, DevOps

PROFESSIONAL EXPERIENCE

University of Tennessee, Knoxville, *Graduate Research Assistant* Aug 2022 - Jul 2026
Designed scalable, reliable, and interpretable AI, deep learning, and LLM workflows; data engineering and fusion for multimodal datasets; worked in a cross-disciplinary team.

ITQAN Analytics and Software Limited, *Developer Intern* Mar 2022 - Jul 2022
Developed Python data engineering pipelines and Power BI dashboards for data visualization.

Monstar Lab Bangladesh, *Frontend Intern* Dec 2019 - Feb 2020
Developed chatbot interfaces using AngularJS and Node.js with senior mentors.